

Emotional Dynamics and Consensus Quality: A Case Study of Hebrew-Wikipedia Mental-Health Talk Pages

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1 Research question

Do emotional expressions in talk-page discussions (aggression, conciliation, exhaustion, contagion) predict the structure of the discussion (dropout vs. convergence) and, through it, the quality of the resulting article version—measured as *stability* (survival of a consensus version without reverts) and *correctness* (expert accuracy ratings)? Do stability and correctness agree?

2 Infrastructure

Code: <https://github.com/nimrodtalmon/Wikipediadiscussions>. Public site with a home-page, a methodology page (`method.html`), and the corpus explorer (article timeline → thread table → reader with discussion tree and outcome strip, implementation switch, replay slider): <https://nimrodtalmon.github.io/Wikipediadiscussions/>. All data is drawn live from the MediaWiki API of Hebrew Wikipedia; raw data is regenerated on demand and not stored in the repository.

3 Pipeline, by module

Working strategy. Build the complete pipeline end-to-end in provisional (v0) form first — that is the current state — and only then rewrite it module by module to a scientific, validated standard. Every definition below is what the code does *today*; exact operational definitions are mirrored, in Hebrew, on the project’s methodology page (`method.html`), so the web application is self-contained.

Each module states what is implemented now and what requires joint revision. The pipeline is deliberately modular: each module is a pure function of the previous one’s output, a module may have several interchangeable implementations (emotion coding has two; quality rating will have two), and the visualizations update from the rebuilt data file. The public site was slimmed accordingly (curation tool now internal; redundant sentiment band removed).

3.1 1. Corpus and data acquisition

Currently implemented: Category crawl (depth 2) from 16 mental-health-adjacent roots on Hebrew Wikipedia: 456 categories, 7,801 articles, of which 2,957 have a talk page of at least 2KB. The top 800 by talk size were ranked by talk-page edit counts (capped at 500 per page), yielding 834 ranked candidates. A curation interface lists them with their category path; selections export

a seed list. A provisional seed of 18 articles (13 clinical core, 5 adjacent charged topics) drives all numbers below.

To revise with Anat: inclusion criteria (clinical vs. adjacent ring; politically charged articles are currently *in*), target corpus size (~ 60 – 100 articles), and the final curated list.

Data acquisition. Currently implemented: For each article: current text of the talk page and its archive subpages, plus full revision metadata (timestamps, users, sizes, comments) of both the talk pages and the article itself, via the MediaWiki API with polite rate-limiting and retries.

To revise with Anat: nothing; purely technical.

3.2 2. Discussion segmentation

Currently implemented: Talk pages are split into threads by section headings; threads are split into comments by signature timestamps and indentation; each comment carries editor, date, indentation depth, and a reply-tree parent (nearest shallower predecessor). Known noise: unsigned fragments and inconsistent indentation conventions.

To revise with Anat: definition of a “participant turn” for the psychological analysis (e.g. merging consecutive fragments by the same editor), and how to treat unsigned comments.

3.3 3. Comment emotion (two implementations)

Currently implemented: Placeholder v0: Hebrew lexicon matching for three categories—aggression, conciliation, frustration/exhaustion—plus neutral; per-comment label and intensity. On the seed corpus (2,670 comments): 175 aggressive, 564 conciliatory, 37 exhausted, 1,894 neutral. Conciliation over-fires on routine politeness; sarcasm and formal hostility are missed. **Implementation B (LLM coding).** Same output schema, produced comment-by-comment in rating sittings and stored by key; the explorer switches between implementations. Coverage is now complete for the rich-thread corpus: all 1,220 comments in the 71 rich threads, coded over six sittings. Corpus distribution — LLM: 208 aggressive, 270 conciliatory, 27 exhausted, 715 neutral; lexicon: 159 / 250 / 25 / 786. Comment-level agreement is 61%. The disagreement is concentrated where it matters: the lexicon misses sarcasm, condescension, status-based dismissal (“you are anonymous, learn the rules”), and polite accusation — most of what aggression looks like in a Wikipedia argument — and over-labels routine thanks as conciliation. Thread-level aggression is strongly bimodal: 40 of 71 rich threads have $\leq 5\%$ aggressive comments (featured-article reviews, naming discussions), while 7 have $\geq 30\%$ (content disputes on gender dysphoria, gaslighting, ADHD, pedophilia, alternative medicine). The lexicon itself now lives in an editable CSV (`data/lexicon.csv`); logic and data are separated.

To revise with Anat: the emotion taxonomy itself (categories, definitions, granularity); coding scheme (per comment vs. per turn); the plan for LLM-based coding of rich threads; a manual coding protocol and sample size for validation (inter-rater agreement, and agreement between manual, lexicon, and LLM codings). Contagion is a derived dynamic (Module module 4), not a per-comment label.

3.4 4. Discussion dynamics

Currently implemented: v0 per-thread measures computed on the module 2 comment sequence with module 3 labels (inheriting both layers’ noise): *emotional contagion* as (a) lag-1 correlation of a valence series (+1 conciliation, -1 aggression, -0.5 exhaustion) and (b) aggression

lift, $P(\text{next aggressive} \mid \text{aggressive})/P(\text{aggressive})$; *hostile exits* — recently-active participants who never speak again after another editor’s aggressive comment; *duels* — the longest strictly alternating run between two editors; *concentration* (top-editor share); and a *convergence proxy* (last three comments contain conciliation and no aggression). Seed-corpus headlines: aggression lift has median 2.1 and exceeds 1 in 12 of 14 measurable threads — aggression begets aggression even under lexicon-level coding; 55 hostile exits corpus-wide; 27 of 71 rich threads end conciliatory; 11 threads contain a duel of ≥ 6 turns. Lag-1 valence correlation is essentially null (median $-.06$), as expected when most comments are labeled neutral.

To revise with Anat: which of these constructs are psychologically meaningful, their operational definitions and thresholds, and what to add (e.g. trajectories over participants rather than comments).

3.5 5. Article stability

Currently implemented: Per article, from full revision history: *identity reverts* (a revision whose content hash equals an earlier revision’s within a 50-edit window, i.e. the page returned exactly to a prior state), summary-based reverts as a secondary signal, edit-war episodes (≥ 3 identity reverts within 48h of one another), and version lifetimes (median and longest inter-edit gap; age of the current version). Seed-corpus headline: pedophilia is the most contested article (24% identity-revert rate, 16 war episodes); ADHD is the most edited (3,155 revisions) but among the calmest (5%).

To revise with Anat: nothing yet; window and war thresholds to be calibrated, and passage-level (rather than whole-page) stability is future work.

3.6 6. Version quality (implementations: LLM, expert; applied to two samples)

Currently implemented: *Definition, independent of any discussion:* given a revision and a passage of interest, an LLM rates the passage 1–7 on accuracy, sourcing, and neutrality with a one-sentence rationale. Which revisions get rated is a sampling decision made in the analysis layer. *Sample A (surviving version):* for each rich thread, the first article revision after the discussion window; a passage (4000 chars) is located by thread-title keywords, falling back to the lead. All 70 ratable rich threads (one lacks any post-discussion revision) were rated in-session by Claude in September 2026; scores are live in the explorer and on the article timelines. Corpus means: accuracy 4.9, sourcing 3.9, neutrality 5.0 (of 7); nine versions scored accuracy ≤ 3 . The pilot already caught a substantive error: the Hebrew gender-dysphoria article’s APA quotation dropped a negation, reversing the original’s meaning (“gender nonconformity is *not* in itself a mental disorder”). Keys are now index-based (unique). Known v0 limits: the rated passage is located by keyword match and often falls back to the article lead, so ratings partly reflect lead quality rather than the disputed passage; and the rater is a general-purpose LLM without a jointly fixed rubric.

To revise with Anat: the rubric (dimensions, anchors) so the LLM prompt and the expert protocol are aligned.

Sample B — version pairs around the pivotal revert

Currently implemented: Rating only the surviving version cannot detect *wrongful reverts* — a good edit reverted by a wrong (possibly aggressive) editor. For each rich thread with an identity revert in its window, the *pivotal revert* is the first such revert; the *kept* version is the revert revision itself and the *discarded* version its predecessor. The largest changed line-block (± 6 lines; line-level diff) is compared and judged: kept better / discarded better (wrongful) / tie / self-revert / unclear.

All 48 pairs on the seed corpus were judged (LLM, in-session): 27 rightful, 3 ties, 7 self-corrections, 9 unclear (the changed block exceeded the extraction window — a known extraction fix), and **2 wrongful**. One wrongful case closes a loop with module 6 (LLM): on the anxiety-disorder article, an anonymous editor removed the clinically imprecise claim that a panic attack “is traumatic”; a registered editor reverted the removal, and module 6 (LLM) independently found the sentence still present years later — a documented instance of a good edit suppressed and an error stabilized.

To revise with Anat: the judgment rubric (what counts as “better”), and inclusion of these pairs in the expert-validation sample.

Implementation: expert rating (not yet built)

Not yet implemented; entirely joint. Expert ratings on a subsample validate (or replace) the module 6 (LLM) proxy: rater recruitment, rubric, unit of rating, sample size, agreement analysis.

To revise with Anat: everything; this is the clinical anchor of the project.

3.7 7. Linking discussions to edits

Currently implemented: v1: time window plus a linking score. MediaWiki provides no structural pointer from a talk-page thread to an edit or passage—threads are free text—so linkage must be inferred. Each thread keeps its v0 time window (one day before its first signed comment to seven days after its last; edit/revert counts; post-thread stability, right-censored). On top of it, a linking score sums five independent signals: an explicit diff/permalink in the thread text (+3; a hard anchor, but present in only 12 of 474 threads on the seed corpus); *BRD ordering*—an identity revert in the three days before the thread opened (+2); editor overlap between thread participants and in-window article editors (+2 if ≥ 2 shared editors, +1 if one); an in-window edit summary citing the talk page (+2); and a thread-title token appearing in an in-window edit summary, exploiting MediaWiki’s automatic `/* section */` summaries (+2). Confidence: high ≥ 5 , medium 3–4, low ≤ 2 . Of the 71 rich threads: 28 high, 25 medium, 18 low.

Causal framing. Wikipedia’s Bold–Revert–Discuss norm means discussions typically *follow* reverts rather than precede them; the correct event model of a dispute is edit $\rightarrow$ revert $\rightarrow$ discussion $\rightarrow$ resolution edits, with the talk page and the article as two faces of one episode. Accordingly, the score identifies shared dispute episodes and never a direction of causation; analyses should treat post-episode stability as the outcome of the episode as a whole.

To revise with Anat: nothing structural; the high/medium/low thresholds can be revisited once a hand-validated sample of linked pairs exists (a natural small task for the joint validation round). A stratified sample of 18 rich threads (six per confidence level) with direct links to each thread and to the article history around its window is ready in `data/validation_sample.csv`; the task is to judge, per thread, whether the in-window edits indeed belong to the same dispute (est. one hour).

4 Status summary

5 Results: v0 smoke test (September 2026)

This section documents a first end-to-end run whose purpose is to verify that the pipeline produces analyzable numbers—*not* to test hypotheses. Every measurement layer is v0 (lexicon emotions, time-window linking, LLM quality proxy, lead-biased passage extraction), so estimates below should be read as plumbing checks.

Module	Status	Joint revision needed	Blocking
1 corpus	candidates ready	yes: criteria + final list	module 1 full run
1 acquisition	done (seed)	no	—
2 segmentation	done, v1	yes: turn definition	module 4
3 emotion	lexicon v0 + LLM (complete, 1,220)	yes: taxonomy + validation	—
4 dynamics	done, v0	yes: constructs	—
5 stability	done, v0	no	—
6 quality (LLM, sample A)	done, v0 (70 rated)	yes: rubric	—
6 quality (LLM, sample B: pairs)	done, v0 (48 judged)	yes: rubric	—
6 quality (expert impl.)	—	yes: protocol	analysis
7 linking	done, v1 (scored)	validation sample	—

Table 1: Pipeline status, September 2026.

Method. Unit: rich threads (≥ 5 signed comments; $n = 71$; 70 with a quality rating). Predictors: shares of aggressive/conciliatory comments per thread. Outcomes: excess in-window reverts (reverts in the thread’s window divided by the article’s baseline revert rate times window length), post-thread stability (log days to next identity revert; right-censored), and the module 6 (LLM) accuracy score. Spearman correlations; Mann–Whitney for aggression-present vs. absent (uncensored only). Code: `src/analysis.py {lex|llm}`; outputs `data/analysis_lex.json`, `data/analysis_llm.json`. The table below is the lexicon run; the LLM run follows.

Test	ρ	p	n
Aggression share vs. excess in-window reverts	−0.10	.42	71
Aggression share vs. post-stability	+0.16	.20	71
Conciliation share vs. post-stability	−0.15	.20	71
Aggression share vs. accuracy	−0.20	.10	70
Post-stability vs. accuracy	−0.04	.77	70
Number of editors vs. post-stability	−0.36	.002	71
Article sentiment balance vs. revert rate	+0.14	.58	18

Table 2: Spearman correlations, v0 smoke test. Post-stability is $\log_{10}(\text{days}+1)$.

Findings. Three observations worth carrying forward. (1) The only robust association is structural: threads with more editors precede less stable versions ($\rho = -.36$)—plausibly confounded by article activity, but a sensible sanity check that the linking layer carries signal. (2) Stability and accuracy are uncorrelated ($\rho = -.04$): if this survives better measurement, it is the paper’s central provocation—Wikipedia can be stably wrong. (3) The emotion associations are directionally sensible (aggression weakly negative for accuracy) but nowhere near significance, as expected given lexicon-level measurement; this is the strongest argument for prioritizing the module 3 taxonomy work.

Re-running the smoke test on the LLM implementation. With emotion coding switched from lexicon to LLM (everything else unchanged), the picture does *not* sharpen; it shifts. Aggression share vs. accuracy weakens ($\rho = -.12$ vs. $-.29$ under the current lexicon); aggression vs. post-stability stays null ($+0.18$). Two conciliation effects appear, both in the direction *opposite* to the naive hypothesis: conciliation share vs. post-stability $\rho = -.23$ ($p = .055$), and threads that end conciliatorily are followed by *shorter* stability (median 42 vs. 101 days, Mann–Whitney $p =$

.06). The likely explanation is confounding by article activity: conciliatory endings are typical of featured-article reviews on the most heavily edited articles (autism, ADHD), which are also reverted soonest. This is the strongest argument yet for the planned normalization against baseline article activity and for article-level clustering before any interpretation. The structural result (editors vs. stability, $\rho = -.36$) is implementation-invariant. Both runs are stored (`data/analysis_lex.json`, `data/analysis_llm.json`).

Dynamics addendum. Adding module 4 measures to the smoke test: hostile-exit share correlates weakly negatively with accuracy ($\rho = -.19$, $p = .13$, same direction as aggression share) and not at all with stability; convergence-ending threads do not show longer post-stability ($p = .29$); lag-1 contagion is uninformative at v0 coding. The one dynamics result that stands out is the within-thread aggression lift (median 2.1), which is a property of sequences rather than a correlate and survives as a descriptive finding.

What must improve before any claim (in priority order). (1) Normalize outcomes against baseline article activity and cluster by article — the LLM re-run shows activity confounding dominates every emotion effect; (2) validate module 3 against a manually coded sample with an agreed taxonomy; (3) passage-level linking in module 7 instead of time windows; (4) survival analysis that honors censoring; (5) a shared rubric for module 6 before expert rating.

6 Status and next steps

The pipeline exists end to end in provisional form: every module has at least one implementation, the emotion module has two (lexicon, LLM — fully coded), the quality module has been applied to two samples (70 surviving versions; 48 revert pairs), and a first analysis has run under both emotion implementations. The public site is self-contained: a methodology page describes each module (purpose, current implementation, technical definition), and the explorer lets a reader switch implementations and replay any article’s or discussion’s history.

The working strategy from here is unchanged: rewrite module by module to a validated standard, in the priority order above, with the joint decisions listed under each module (corpus criteria, turn definition, emotion taxonomy, dynamics constructs, quality rubric) taken at the kickoff with the psychology side. Nothing in the results section should be cited until items (1) and (2) are done.